

5.4 Organic chemistry – arenes, nitrogen compounds and synthesis

Knowledge of the common uses of organic compounds mentioned in this topic is expected.

Students will be assessed on their ability to:

1 Arenes: benzene

- a use thermochemical, x-ray diffraction and infrared data as evidence for the structure and stability of the benzene ring

Students may represent the structure of benzene as



as appropriate in equations and mechanisms

- b describe the following reactions of benzene, limited to:
- i combustion to form a smoky flame
- treatment with:
- ii bromine
 - iii concentrated nitric and sulfuric acids
 - iv fuming sulfuric acid
 - v halogenoalkanes and acyl chlorides with aluminium chloride as catalyst (Friedel-Crafts reaction)
 - vi addition reactions with hydrogen
- c describe the mechanism of the electrophilic substitution reactions of benzene in halogenation, nitration and Friedel-Crafts reactions including the formation of the electrophile
- d carry out the reactions in 5.4.1b where appropriate (using methylbenzene or methoxybenzene)
- e carry out the reaction of phenol with bromine water and dilute nitric acid and use these results to illustrate the activation of the benzene ring.

2 Organic nitrogen compounds: amines, amides, amino acids and proteins

- a give examples of:
 - i molecules that contain amine and amide functional groups
 - ii amino acids
- b describe and carry out, where appropriate (using butylamine and phenylamine), reactions to investigate the typical behaviour of primary amines. This will be limited to:
 - i characteristic smell
 - ii miscibility with water as a result of hydrogen bonding and the alkaline nature of the resulting solution
 - iii formation of salts
 - iv complex ion formation with copper(II) ions
 - v treatment with ethanoyl chloride and halogenoalkanes, e.g. making paracetamol
- c describe the reduction of aromatic nitro-compounds using tin and concentrated hydrochloric acid to form amines
- d describe and carry out, where appropriate, the reaction of aromatic amines with nitrous acid to form benzenediazonium ions followed by a coupling reaction with phenol to form a dye
- e recall the synthesis of amides using acyl chlorides
- f describe:
 - i condensation polymerisation for the formation of polyesters such as terylene and polyamides such as nylon and Kevlar
 - ii addition polymerisation including poly(propenamide) and poly(ethenol)
- g draw the structural formulae of the repeat units of the polymers in 5.4.2f
- h comment on the physical properties of polyamides and the solubility in water of the addition polymer poly(ethenol) in terms of hydrogen bonding, e.g. soluble laundry bags or liquid detergent capsules (liquitabs)

- i describe and carry out, where appropriate, experiments to investigate the characteristic behaviour of amino acids. This is limited to:
 - i acidity and basicity and the formation of zwitterions
 - ii separation and identification by chromatography
 - iii effect of aqueous solutions on plane-polarised monochromatic light
 - iv formation of peptide groups in proteins by condensation polymerisation
 - v reaction with ninhydrin.

3 Organic synthesis

- a give examples to illustrate the importance of organic synthesis in research for the production of useful products
- b explain why sensitive methods of chemical analysis are important when planning and monitoring organic syntheses
- c deduce the empirical formulae, molecular formulae and structural formulae from data drawn from combustion analysis, elemental percentage composition, characteristic reactions of functional groups, infrared spectra, mass spectra and nuclear magnetic resonance
- d use knowledge of organic chemistry contained in this specifications to solve problems such as:
 - i predicting the properties of unfamiliar compounds containing one or more of the functional groups included in the specification, and explain these predictions
 - ii planning reaction schemes of up to four steps, recalling familiar reactions and using unfamiliar reactions given sufficient information
 - iii selecting suitable practical procedures for carrying out reactions involving compounds with functional groups included in the specification
 - iv identifying appropriate control measures to reduce risk during a synthesis based upon data of hazards
 - v understanding why, in the synthesis of stereo-specific drugs, it is important to understand the mechanism of the reaction and how this can help to plan the synthesis
- e explain why the pharmaceutical industry has adopted combinatorial chemistry in drug research, including passing reactants over reagents on polymer supports

- f describe and carry out, where appropriate, the preparation of a compound, e.g. cholesteryl benzoate (a liquid crystal) and of methyl 3-nitrobenzoate, requiring some of the following techniques:
- i refluxing
 - ii purification by washing, e.g. with water and sodium carbonate solution
 - iii solvent extraction
 - iv recrystallisation
 - v drying
 - vi distillation
 - vii steam distillation
 - viii melting temperature determination
 - ix boiling temperature determination.